

The Effect of Contamination on Contamination Limit Regulation

US Coal Mining and Drinking Water Utilities

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September 2026

Self-reported compliance and US drinking water monitoring

- Regulatory models requiring entities to self-report compliance:
 - ▶ Personal income taxes; Healthcare HIPAA violation disclosure; Energy firms self-report anti-competitiveness/negligence to the Federal Energy Regulatory Commission; Public companies' SOX self-report financial statement accuracy; Toxic Release Inventories; Drinking water utilities.
- Incentive to shirk reporting. Consider drinking water utilities federal water quality regulation:
 - ▶ No self-report $\Rightarrow$ no contamination exceedance violation issued.
 - ▶ No self-report saves self-reporting costs (Oleckno, 1982).
- **Incentives rise with contamination**
 - ▶ $\uparrow$ contamination $\rightarrow \uparrow$ testing requirements ($\uparrow$ **reporting costs**)
 - ▶ $\uparrow$ contamination $\rightarrow \uparrow$ probability contamination violation ($\uparrow$ **need to conceal**).
- Firms under-report when reporting is most valuable to the regulator.

Self-reported compliance of drinking water quality

- Drinking water utilities provide treated water to 85% of US households.
 - ▶ Utilities source from groundwater and/or surface water, serve twenty five to millions of customers, publicly or privately owned.
- Public health implications of self-report/under-reporting.
- 9 to 34 million US households obtain water from a utility violating a health standard (Allaire, Wu, and Lall, 2018).
- Whether it is self-reporting costs or concealment has different policy implications for improving reporting.

Research question

What happens to firms and regulators when contamination of a regulated output rises exogenously, under self-reported compliance?

- Does contamination reduce firm self-reporting?
- Is the reduction concealment of exceedances, or avoidance of reporting costs?
- Does contamination increase regulator enforcement?

Coal mining contaminates water and is not gone

- Coal supplied 16% of US electricity in 2025 (International Energy Agency (2025)).
- Federal policy is expanding mining access and delaying plant retirements (U.S. President (Trump) (2025); Daly (2025)).
- **Acid mine drainage:** exposed sulfide rock oxidizes, forms sulfuric acid, dissolves metal ions and sulfates, which travel into surface and ground water (Skousen, Ziemkiewicz, and McDonald (2019)).
- **Coal wash slurry:** impurities stored in retaining ponds seep through porous rock, or the ponds fail.
- Both increase with the **sulfur content** of the coal seam.
- Contaminants: **arsenic, nitrate, barium, selenium** — inorganic chemicals with stable utility MCL's 1985–2005.
- Costly to remove: replacing one small system's arsenic filtration cost \$192,000 plus \$60/day to run (US EPA, 2011).

What I do

- Setting: drinking water utilities downstream of coal mines, 1985–2005.
 - ▶ Contamination is **upstream externality** — the firm cannot reduce by cutting output (Lori Snyder Bennear, 2005).
 - ▶ Separation aids identifying the effect of contamination.
- Adapt Douglas and Wiggins (2015) instrument for coal production to address two sources of endogeneity:
 - ▶ Coal mines are near coal seams and transportation that independently predict water quality and utility characteristics.
 - ▶ Regulators block upstream mining for vulnerable utilities.
- Contributions:
 - ▶ First causal estimates of effect of coal mining on **treated** drinking water at US utilities.
 - ▶ First evidence that **the cost of self-reporting**, not only concealment, causes environmental under-reporting (Naylor, 2024; Zou, 2021; Lori S. Bennear, Jessoe, and Olmstead, 2009).
 - ▶ Causal evidence of effect of **contamination itself** on contamination control effectiveness.

What I find

- $\uparrow$ mining $\rightarrow$ $\uparrow$ contaminant concentrations. 10M cumulative tons upstream raises mean nitrate by 11.0% of its sample mean (***) and selenium by 8.5% (**); arsenic and barium show no significant change. All stay below the MCL.
- $\uparrow$ mining $\rightarrow$ $\uparrow$ **under-reporting**. One more upstream mine raises MR violations by 3–5 pp, against base rates of roughly 4–6%.
- $\uparrow$ mining $\rightarrow$ **no significant change** in MCL violations for nitrates or IOCs; a small but significant arsenic increase (+0.08 to +0.22 pp).
- $\uparrow$ mining $\rightarrow$ $\uparrow$ **inspection visits** (+2.1 pp); sanitary visits, enforcement visits, and formal enforcement lose significance once state $\times$ year fixed effects are added (formal enforcement: -3.4 pp with utility and year fixed effects alone).
- **Self-reporting costs, not concealment:**
 - ▶ when nitrate readings cross 50% of the MCL and monitoring requirements step up — reporting cost up — increase under-reporting.
 - ▶ Sanitary inspections at under-reporting utilities do not discover concealed violations.
 - ▶ Contaminant concentrations stay below MCL.

Self-reporting model

Institutional setting

- Maximum contaminant level (MCL)
- Monitoring and reporting violation (MR)
- States are Safe Drinking Water Act (SDWA) enforcers, have wide discretion over enforcement.
- Enforcement while remedial before punitive is still costly:
 - ▶ Informal (letters, notices, compliance plans)
 - ▶ Formal (consent decrees, fines, prosecution)
 - ▶ Visits (sanitary, sampling, technical assistance)
- MR violations are punished less severely than MCL violations.

Model: the utility's decisions to self-report or under-report

Utility draws compliance cost type $\theta \sim F(\cdot; m)$ and self-reporting cost $c \sim G(\cdot; m)$, both worsening in upstream contamination m .

- Chooses negligence a , unobserved by regulator, with benefit $\theta b(a)$, $b' > 0$, $b'' < 0$.
- Each of N required tests would reveal an exceedance with probability $q(a)$, $q' \geq 0$.
- Penalties: r for a self-reported exceedance, t for an MR violation, $s > r$ if an inspection (probability p) uncovers a hidden exceedance.

$$\underbrace{c + q(a)r}_{\text{report}} \leq \underbrace{t + p q(a)s}_{\text{under report}} \implies \text{report iff } c \leq \hat{c}(a) \equiv t - (r - ps) q(a).$$

- A system with **zero** exceedance risk still stops reporting once $c > t$.
- Assume $ps < r$ to give utility a reason to conceal.

Model: why the utility under reports

Proposition 2

A type- θ utility under-reports if and only if its cost draw exceeds this cutoff, $c > c^*(\theta)$, so the share of utilities under-reporting and receiving an MR violation is:

$$MR(m) = \int_0^{\bar{\theta}} [1 - G(c^*(\theta); m)] f(\theta; m) d\theta \equiv \int_0^{\bar{\theta}} h(\theta, m) f(\theta; m) d\theta.$$

Differentiating (9) splits into two channels:

$$\frac{\partial MR}{\partial m} = \underbrace{\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial m} f(\theta; m) d\theta}_{\text{Channel A: cost-distribution shift}} + \underbrace{\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m} d\theta}_{\text{Channel B: type-distribution shift}}.$$

Data

- **Violations, enforcement, visits** — SDWIS/ECHO, universe of utilities 1985–2005. Binary by utility $\times$ year; days-in-year as a robustness measure.
 - ▶ Under-inclusive but accurate: 95% of the violations SDWIS holds are correctly reported (US EPA, 2000).
- **Contaminant concentrations** — EPA Six-Year Review round 2, 1998–2005. Cleaned following Ravalli et al. (2022); non-detects set to $MDL/\sqrt{2}$.
- **Mines** — MSHA locations merged to EIA annual production, mine level.
- **Coal sulfur %** — USGS COALQUAL samples, averaged within 20 km of each watershed.
- **Hydrology** — NHDPlus HUC12 watershed boundaries and inflow/outflow.
- **Sample:** Utilities with a coal mine **within two watersheds upstream** of their intake and no coal mine colocated with their intake. 10,641 utility-years, 565 utilities.

Matching intakes to the watershed above them

- HUC12 is the finest watershed mapped for the continental US (>87,000 units).
- Exposure = mines and tons coal in the HUC12s **within two watersheds upstream** of the utility's intakes.
- Intakes **colocated** with a mine are excluded.
- Multiple intakes: sum production, average sulfur.



MR violations are common; MCL violations barely exist

Panel A: Any violation in year

Contaminant	MR		MCL	
	%	N	%	N
Nitrates	5.68	612	0.11	12
Arsenic	3.73	402	0.05	5
Inorganic chemicals	4.31	465	0.30	32

Panel B: Days in year

Contaminant	MR				MCL			
	Mean	SD	P90	P99	Mean	SD	P90	P99
Nitrates	14.54	65.35	0.00	365.00	0.27	9.19	0.00	0.00
Arsenic	9.98	54.32	0.00	365.00	0.08	3.83	0.00	0.00
Inorganic chemicals	11.53	58.47	0.00	365.00	0.71	15.08	0.00	0.00

Nitrate MR 5.7%, arsenic 3.7%, IOC 4.3% of utility-years. MCL violations: 12, 5, and 32 events across 1985–2005.

Mining exposure in the sample

Variable	Mean	SD	P90	P99
Number of mines upstream	1.12	2.27	4.00	11.00
Cumul. upstream coal prod. (10M ST) since 1985	0.32	1.00	0.68	5.23
Percentage of coal weight as sulfur	1.11	1.13	2.96	4.01

On average 1.1 mines upstream and 1.1% sulfur by weight; cumulative upstream production reaches 5.2 (10M ST) at the 99th percentile.

Does coal mining contaminate utility water?

Design: cumulative production, measured concentrations

$$Concen_{ct} = \beta \cdot \sum_{\tau=1985}^t \text{UpstreamCoalProd}_{c\tau} + \rho_c + HUC02_c \cdot \tau_t + \epsilon_{ct}$$

- $Concen_{ct}$: mean annual concentration at utility c , year t (SYR2, 1998–2005).
- β : effect of 10 million additional cumulative short tons mined upstream since 1985.
- ρ_c : utility fixed effects. $HUC02_c \cdot \tau_t$: HUC02-specific year effects, absorbing watershed-wide industrial trends.
- SEs clustered at the utility level.
- Continuous treatment, **no control**: identification comes from relative exposure.
- Assumption: utility with large upstream production changes would have trended like those with small ones. (balance test on utility characteristics $\rightarrow$)

Mining raises contaminant levels — but not above the limit

	Arsenic (1)	Nitrate (2)	Barium (3)	Selenium (4)
Panel A: Within one HUC12 upstream				
Cumul. upstream coal prod. (10M ST)	0.0022*** (0.0004)	0.0278 (0.1021)	0.0213** (0.0097)	0.0031** (0.0015)
Panel B: Within two HUC12 upstream				
Cumul. upstream coal prod. (10M ST)	0.0001 (0.0001)	0.0883*** (0.0257)	0.0136 (0.0112)	0.0004** (0.0002)
Observations	366	472	353	350

Notes: All specifications include utility and HUC02 $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Nitrate +0.09 mg/L per 10M tons (***): 11.0% of its mean. Selenium +0.0004 mg/L (**): 8.5% of its mean. Arsenic and barium show no significant change. **Every reading in the sample stays under its MCL.** (effect across choice of upstream watershed radius $\rightarrow$)

Do utilities keep self-reporting?

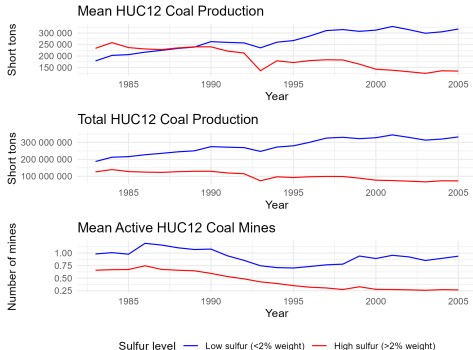
The Acid Rain Program moved coal, not for water reasons

- Adapt Douglas and Wiggins (2015) ARP instrument for coal production to address two sources of endogeneity of an OLS of mining on utility behavior and characteristics:

- ▶ Coal mines are near coal seams and transportation that independently predict water quality and utility characteristics.
- ▶ Regulators block upstream mining for vulnerable utilities.

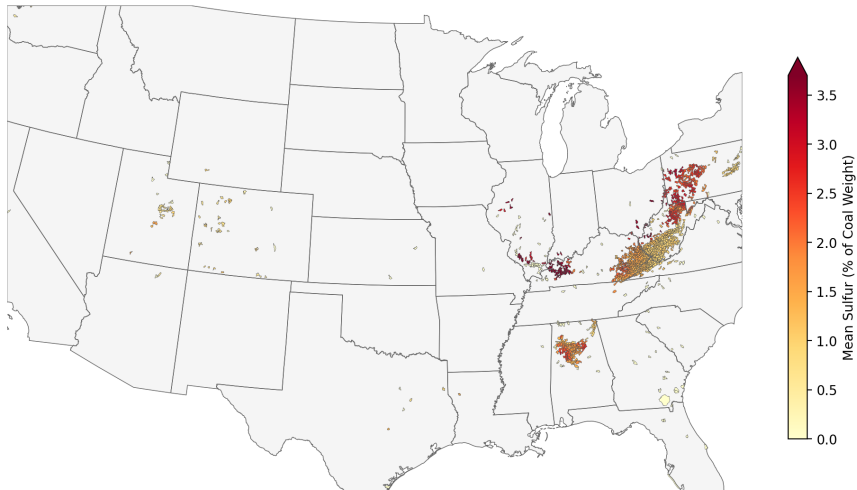
- 1990 Clean Air Act amendments cap SO_2 from power plants; Phase I deadline **1995**.

- Power plants were 90% of coal

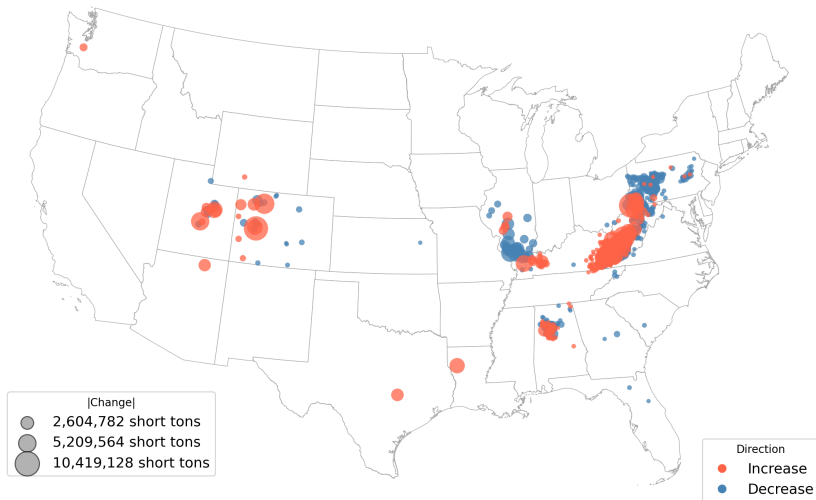


Watershed coal sulfur content

HUC12 watersheds containing coal mines, by mean coal sulfur content



Where coal production moved, 1985–2005



IV design: the ARP as a shifter of where coal is mined

First stage:

$$\text{NumMines}_{ct} = \alpha_0 + \alpha_1 \text{NumFacility}_{ct} + \alpha_2 \text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t \\ + Y_t + C_c + \nu_{ct}$$

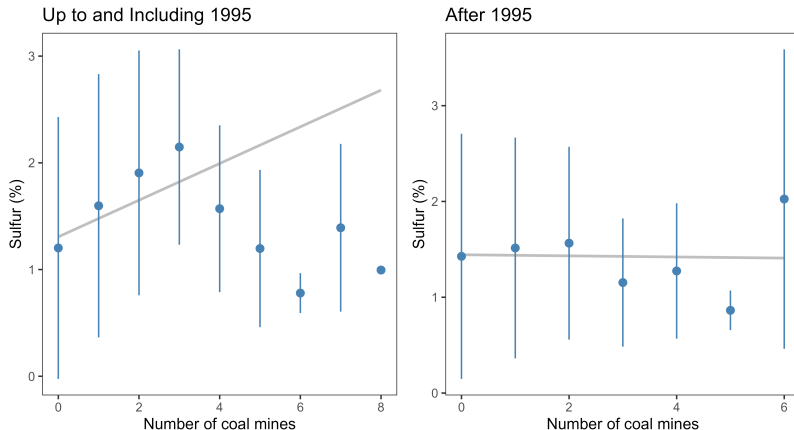
Second stage:

$$\text{Violation}_{vct} = \beta_0 + \beta_1 \text{NumFacility}_{ct} + \gamma \widehat{\text{NumMines}}_{ct} + Y_t + C_c + \epsilon_{ct}$$

- $\text{Post95}_t = \mathbf{1}\{t > 1995\}$. Sulfur is time-invariant, so C_c absorbs its level.
- Outcome: 1 if utility c had a violation of type v at any point in year t . Coefficients are in percentage points.
- SEs clustered at the utility level.

Shift in location of coal production w.r.t. sulfur

Descriptive evidence of first-stage relevance



Sample: HUC12 watersheds directly upstream of drinking water utility water intakes 1985-2005.

Points show the mean sulfur as a percent of coal weight within each coal-mine-count bin; ranges show ± 1 standard deviation within bin.

First stage: high sulfur watersheds lost their mines

	Upstream coal mines (sum) (1)
Post-1995 $\times$ Upstream sulfur %	-0.4933*** (0.0703)
F-test (1st stage, clustered)	49.20
Utility fixed effects	✓
State $\times$ year fixed effects	✓
Observations	10,641
R ²	0.84999
Within R ²	0.05289

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(first-stage strength across choice of upstream watershed radius $\rightarrow$)

Instrument validity

- **Independence.** The ARP is national clean-*air* law. Its scope, stringency, and 1995 date were set by national energy and emissions politics, not by any downstream utility's water quality or its influence over mine siting. Time-invariant sulfur exposure is absorbed by utility fixed effects.
- **Exclusion.** Main exclusion test is to look at effect of mining downstream of utility on utility behavior. (MR violations $\rightarrow$), (visits $\rightarrow$), (enforcement $\rightarrow$) — all placebo estimates on utilities upstream of the mine. The threat is a post-1995 SDWA change hitting high- and low-sulfur utilities differently. National rule changes are absorbed by year effects. Directly testable: regress the number of intake facilities — the observable proxy for the model's negligence type — on the instrument. The coefficient is 0.05–0.06 (s.e. ≈ 0.04), with and without a balanced panel, and not statistically significant. (full balance table $\rightarrow$)
- **Monotonicity.** A defier is a high-sulfur watershed that expanded *because* its coal was high sulfur. Scrubber adoption costs more likely leading to lost market share rather than gain; low-cost high-sulfur regions that expand anyway are never-takers.

Mining makes utilities stop reporting

	Monitoring and reporting (MR) violation					
	Nitrates (1)	(2)	Arsenic (3)	(4)	Inorganic chemicals (5)	(6)
OLS						
Upstream coal mines (sum)	0.68** (0.33)	0.55 (0.34)	0.62** (0.31)	0.42 (0.33)	0.49 (0.32)	0.19 (0.34)
2SLS						
Upstream coal mines (sum)	5.12*** (1.70)	3.94** (1.75)	4.85*** (1.57)	3.78** (1.55)	4.66*** (1.62)	3.06* (1.66)
RF						
Post-1995 × Upstream sulfur %	-2.12*** (0.66)	-1.95** (0.83)	-2.01*** (0.60)	-1.87** (0.73)	-1.93*** (0.63)	-1.51* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

One more upstream mine raises MR violations by 3.9–5.1 pp for nitrates, 3.8–4.9 pp for arsenic, and 3.1–4.7 pp for IOC (across utility-year and utility-year-by-state-year fixed effects), against base rates of 5.7%, 3.7%, and 4.3%. OLS estimates are far smaller — consistent with negative omitted variable bias, as expected if utilities that can keep mines out are also better at compliance. (effect across choice of upstream watershed radius →)

Little record of exceedances

	Maximum contaminant level (MCL) violation					
	Nitrates (1)	(2)	Arsenic (3)	(4)	Inorganic chemicals (5)	(6)
OLS						
Upstream coal mines (sum)	-0.01 (0.01)	-0.00 (0.02)	-0.02 (0.04)	0.01 (0.04)	0.05 (0.07)	0.05 (0.07)
2SLS						
Upstream coal mines (sum)	-0.09 (0.07)	-0.11 (0.17)	0.08* (0.04)	0.22** (0.10)	-0.16 (0.28)	-0.20 (0.26)
RF						
Post-1995 × Upstream sulfur %	0.04 (0.03)	0.05 (0.09)	-0.03** (0.02)	-0.11** (0.05)	0.07 (0.11)	0.10 (0.13)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Nitrate and IOC MCL effects are statistically indistinguishable from zero. Arsenic rises by a small but significant +0.08 to +0.22 pp. This is not strong evidence the water stayed clean: without a self-report, the regulator cannot issue an MCL violation at all, and the arsenic result shows an exceedance can still surface. It is also the wrong prediction for concealment — hiding is worth paying for when an exceedance is at stake, and measured concentrations are almost always below that threshold.

Reporting cost, not concealment: Corollary 1 in the data

- Nitrate rules require **more frequent monitoring** once a reading exceeds 50% of the MCL — a discrete jump in c , with no jump in the penalty for under-reporting and no exceedance yet: a reading at half the limit is still a reading under the limit.
- Exceedance risk is held fixed (Channel B = 0), reporting cost increases (Channel A $\geq$ 0).

	Nitrate MR (1-yr) (1)	Nitrate MR (6-mon) (2)
Concen. > 50% MCL	58.9341** (24.5521)	26.0695** (10.5051)
Mean concen. (z-score)	1.2906 (1.4648)	0.0795 (0.9057)
Observations	835	835

Notes: All specifications include utility and year fixed effects. SEs clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Crossing 50% of the nitrate MCL raises the probability of failing to report within six months by 26 pp, and within a year by 59 pp. Descriptive, not causal — read the sign not magnitude. Utilities under-report when self-reporting gets expensive, not hiding. 29 / 56

How do regulators respond?

Enforcement does not counteract under-reporting

	Informal		Any enforcement Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	-0.15 (0.42)	-0.06 (0.44)	-1.00*** (0.21)	-0.12 (0.19)	0.33 (0.45)	-0.07 (0.48)
2SLS						
Upstream coal mines (sum)	-3.06 (2.15)	-0.71 (2.10)	-3.37*** (0.95)	-1.08 (0.86)	2.37 (2.11)	-0.42 (2.25)
RF						
Post-1995 × Upstream sulfur %	1.26 (0.84)	0.35 (1.03)	1.39*** (0.40)	0.53 (0.41)	-0.98 (0.85)	0.21 (1.11)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

One more upstream mine reduces formal enforcement by 3.4 pp with utility and year fixed effects alone (significant), but the effect loses significance once state × year fixed effects are added; informal enforcement is not significant in either specification. (effect across choice of upstream watershed radius →)

The regulator increases inspections, not enforcement

	Sanitary (1)	Technical assistance (2)	Enforcement (3)	Sample collection (4)	Inspection (5)
OLS					
Upstream coal mines (sum)	0.28 (0.29)	0.10 (0.20)	0.04 (0.11)	-0.07 (0.05)	0.06 (0.13)
2SLS					
Upstream coal mines (sum)	0.78 (1.20)	0.67 (0.95)	1.07 (0.81)	0.48 (0.50)	2.13** (1.07)
RF					
Post-1995 × Upstream sulfur %	-0.39 (0.59)	-0.33 (0.47)	-0.53 (0.39)	-0.24 (0.24)	-1.05** (0.49)
Utility fixed effects	✓	✓	✓	✓	✓
Year fixed effects					
State × year fixed effects	✓	✓	✓	✓	✓
Utilities	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Only inspection visits rise significantly (+2.1 pp); sanitary and enforcement visits are not statistically distinguishable from zero. Regulators do not systematically step up the visit types tied to sanctions. Either there are no hidden exceedances, or visits do not find them (Mi, Zhang, and Liu (2026)); I cannot separate the two, but concealment is unlikely given the MCL results. Either way the penalty for under-reporting does not increase, which Proposition 3 says would work. (effect across choice of upstream watershed radius →)

Conclusion

- Exogenous contamination raises the cost of self-reporting, and self-reporting falls — **even where the quality standard is never breached.**
- The evidence points to cost avoidance, not concealment. Corollary 1 makes the two separable: with exceedance risk held fixed, only the cost channel can move under-reporting — I find flat exceedance and increasing self-reporting costs.
- Enforcement response is mixed and fixed-effects-sensitive: formal action falls with baseline fixed effects but loses significance once state $\times$ year effects are added, and only inspection visits rise significantly — the regulator's response does not robustly keep pace with contamination.
- **Policy.** Tightening MCLs does nothing if nobody tests. Self-reporting penalties are a lever to reduce under-reporting.
- Open question: if regulator monitoring is more efficient than self-reporting.

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Appendix

Appendix: combined MR and MCL violations

	Nitrates		Any violation Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.67** (0.33)	0.54 (0.34)	0.60* (0.31)	0.43 (0.34)	0.56* (0.31)	0.28 (0.33)
2SLS						
Upstream coal mines (sum)	5.06*** (1.70)	3.89** (1.75)	4.94*** (1.57)	4.00** (1.56)	4.56*** (1.64)	3.09* (1.67)
RF						
Post-1995 × Upstream sulfur %	-2.09*** (0.66)	-1.92** (0.83)	-2.04*** (0.60)	-1.97*** (0.73)	-1.89*** (0.64)	-1.53* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The combined effect mirrors the MR effect: nitrate 3.9–5.1 pp, arsenic 4.0–4.9 pp, IOC 3.1–4.6 pp across the two fixed-effects specifications.

Appendix: instrument balance on intake facilities

	Number of Intake Facilities	
	(1)	(2)
Upstream sulfur		0.9942*** (0.3834)
Post-1995 × Upstream sulfur	0.0509 (0.0386)	0.0563 (0.0376)
Utility fixed effects	✓	
Year fixed effects		✓
State × year fixed effects	✓	
Balanced panel		✓
Utilities	565	447
Observations	10,641	9,387

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

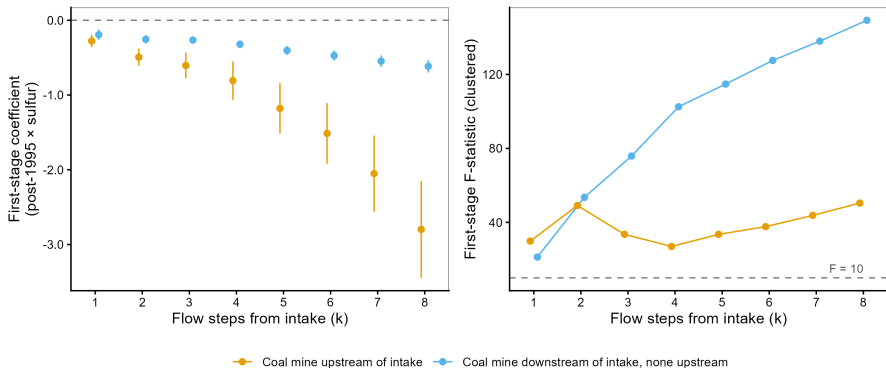
Appendix: balance test on utility characteristics

	Cumul. upstream coal prod. (10M ST)		Any upstream coal mining	
	(1)	(2)	(3)	(4)
Constant	0.2504 (0.2532)		-0.1596 (0.1198)	
Log backcast pop. 1997	0.1091 (0.0750)	0.1512 (0.0938)	0.0159 (0.0126)	0.0208 (0.0144)
Log pop. served (SYR2)	-0.0934 (0.0666)	-0.1113 (0.0956)	0.0060 (0.0259)	-0.0301 (0.0334)
Number of intake facilities	0.0089 (0.0569)	-0.0324 (0.0744)	0.0680* (0.0359)	0.0721* (0.0406)
Surface water source	0.1225 (0.1643)	-0.3902 (0.3296)	0.1210 (0.1307)	0.0521 (0.1601)
Public ownership	-0.1554 (0.2551)	-0.0599 (0.2240)	0.2084** (0.0911)	0.1048 (0.1098)
Wholesaler	-0.3780* (0.2041)	-0.7465 (0.9141)	-0.2164 (0.1340)	-0.2212 (0.1683)
Source water protection	-0.4549 (0.2749)	-0.4385 (0.4305)	-0.2807* (0.1521)	-0.1460 (0.1930)
Joint <i>F</i> -test (all covariates)	2.33	0.96	8.96	2.46
<i>p</i> -value	0.029	0.461	0.000	0.023
Observations	122	120	122	120
State fixed effects		✓		✓

Notes: Heteroskedasticity-robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

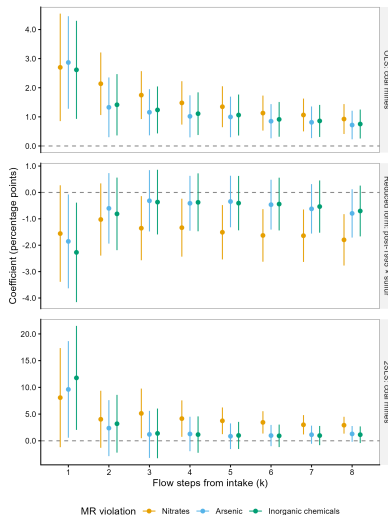
Observable characteristics jointly predict continuous upstream production only without state fixed effects ($p=0.029$) and lose significance once state fixed effects are added ($p=0.461$), as in the main specification. The joint test on the binary presence of upstream mining stays significant with or without state fixed effects.

Appendix: first stage across choice of upstream watershed radius



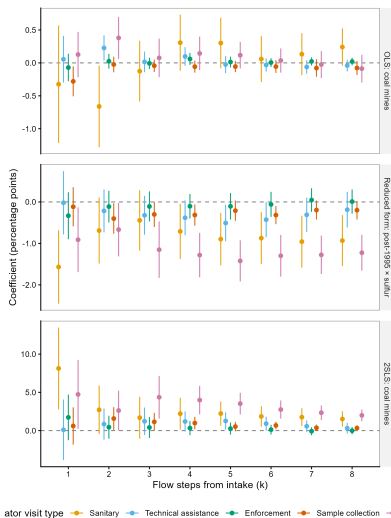
The first-stage coefficient and F-statistic for utilities downstream of a mine stay well above conventional weak-instrument thresholds as the upstream watershed radius k varies from one to eight flow steps; the two-watershed radius used in the main specification is a conservative choice.

Appendix: exclusion placebo, MR violations across k



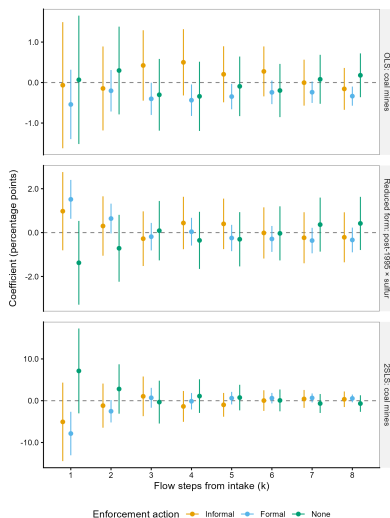
For utilities upstream of a mine — where mining cannot reach the intake hydrologically — the reduced-form and 2SLS estimates on MR violations are small and not statistically distinguishable from zero at every choice of k .

Appendix: exclusion placebo, visits across k



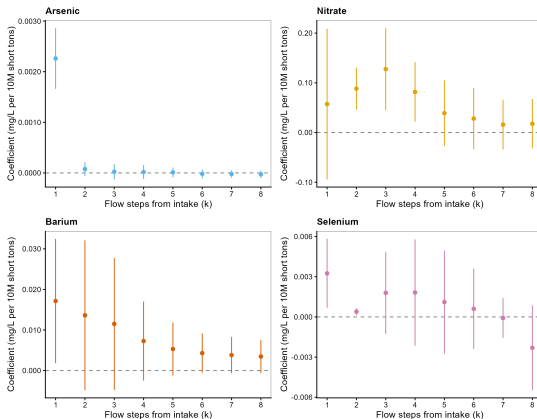
Visit types show no consistent pattern for utilities upstream of a mine at any choice of k , unlike the significant rise in inspection visits found downstream of mining.

Appendix: exclusion placebo, enforcement across k



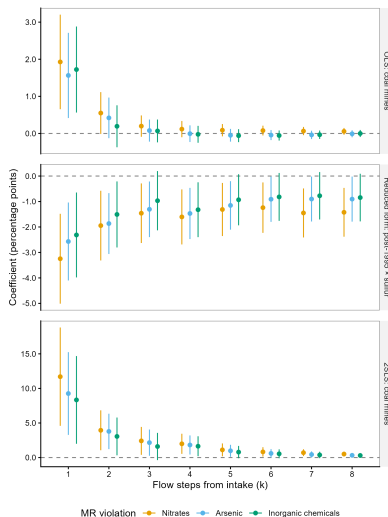
Enforcement actions show no consistent pattern for utilities upstream of a mine at any choice of k , supporting the exclusion restriction.

Appendix: concentration effects across choice of upstream watershed radius



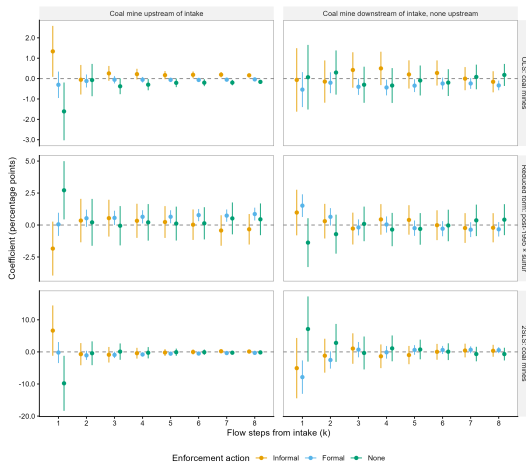
The nitrate and selenium effects are largest at short radii and shrink as k grows; arsenic and barium remain indistinguishable from zero throughout, consistent with the main specification.

Appendix: MR violation effect across choice of upstream watershed radius



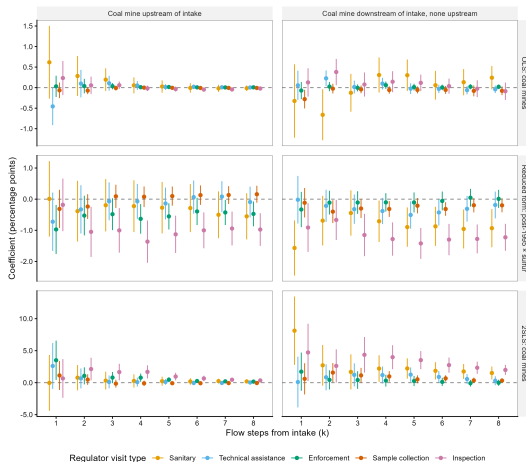
The 2SLS effect on MR violations is largest at $k = 1$ and declines smoothly as the upstream radius widens, but stays positive and significant across all choices of k

Appendix: enforcement effect across choice of upstream watershed radius



The formal-enforcement estimate is not stable in sign or significance across choices of k , consistent with the fixed-effects sensitivity found in the main specification.

Appendix: visit effect across choice of upstream watershed radius



The rise in inspection visits downstream of mining is present across choices of k , while sanitary and enforcement visits remain insignificant throughout.

Appendix: concentration summary statistics

Variable	MCL	Mean	Max	Std. Dev.	N	Near MCL
Panel A: Full sample						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0028	0.0110	0.0023	416	0.0%
Nitrate (mg/L)	10.0 mg/L	0.8020	5.5400	0.8738	488	0.4%
Barium (mg/L)	2.000 mg/L	0.0740	0.7071	0.0659	407	0.0%
Selenium (mg/L)	0.050 mg/L	0.0047	0.0354	0.0049	405	0.5%
Panel B: Utility exposed to above median cumulative tons of coal extraction						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0032	0.0100	0.0026	205	0.0%
Nitrate (mg/L)	10.0 mg/L	0.8254	5.5400	0.8748	260	0.4%
Barium (mg/L)	2.000 mg/L	0.0811	0.3320	0.0569	196	0.0%
Selenium (mg/L)	0.050 mg/L	0.0038	0.0354	0.0043	196	1.0%

Appendix: who appears in the Six-Year Review

Panel A: 1997–2005

	Utility appears in SYR2		Diff.	<i>p</i> -value
	Yes (1)	No (2)		
Ever MR violation	32.2%	18.0%	14.2%***	0.000
Ever MCL violation	1.7%	1.2%	0.5%	0.619
Mean annual violations (count, 0–6)	0.09	0.09	-0.00	0.874

Panel B: 1985–2005

	Utility appears in SYR2		Diff.	<i>p</i> -value
	Yes (1)	No (2)		
Ever MR violation	51.3%	44.1%	7.2%*	0.090
Ever MCL violation	5.5%	3.6%	2.0%	0.277
Mean annual violations (count, 0–6)	0.13	0.20	-0.07**	0.011

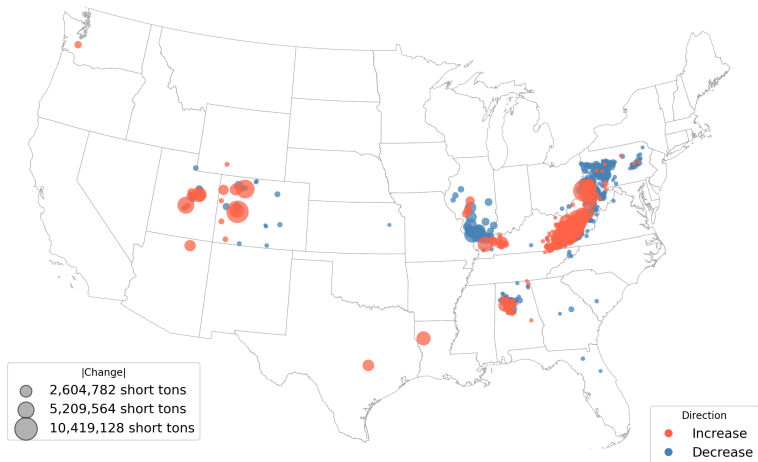
Utilities with SYR2 readings are 7–14 pp more likely to ever incur an MR violation. The concentration sample over-represents the more negligent utilities, so it is not selected toward clean systems.

Appendix: enforcement and visits in the panel

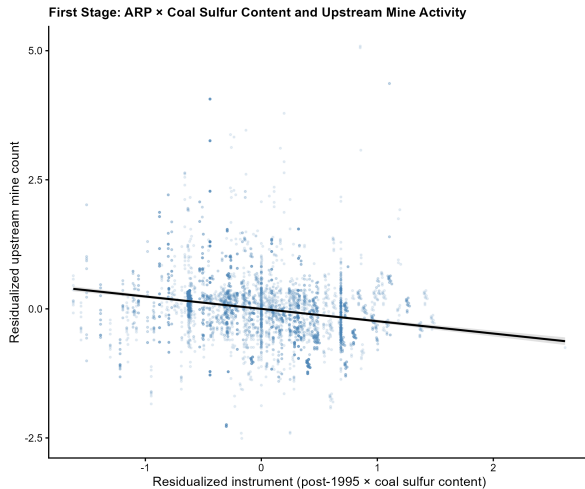
	Whole panel		During MR year		During MCL year	
	%	N	%	N	%	N
Panel A: Enforcement type						
Formal	3.00	323	9.41	70	6.12	3
Informal	18.24	1,965	66.67	496	59.18	29
Any	20.75	2,235	70.70	526	67.35	33
Panel B: Visit type						
Sanitary	13.59	1,464	13.17	98	10.20	5
Technical assistance	2.29	247	4.70	35	2.04	1
Enforcement	1.05	113	2.02	15	4.08	2
Sample collection	0.78	84	0.54	4	0.00	0
Inspection	3.43	369	5.65	42	2.04	1

Enforcement responds to violations — any enforcement rises from 21% overall to 71% in MR years — but sanitary visits are no more likely in MR years than in any other year.

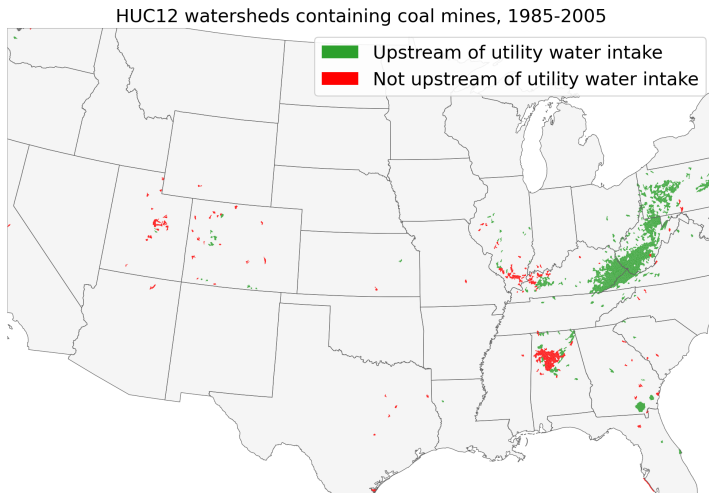
Appendix: where coal production moved, 1985–2005



Appendix: residualized first stage



Appendix: sample watersheds



Appendix: Corollary 1, statement and proof

Write the share of types under-reporting as $h(\theta, m) = 1 - G(c^*(\theta); m)$, with cutoff $c^*(\theta) = t - (r - ps)q(a_{SR}(\theta))$, so that

$$\frac{\partial MR}{\partial m} = \underbrace{\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial m} f(\theta; m) d\theta}_{\text{A: cost}} + \underbrace{\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m} d\theta}_{\text{B: composition}}.$$

Corollary 1. If $q(a_{SR}(\theta)) \equiv \bar{q}$ on $(0, \bar{\theta})$, then $\partial MR / \partial m = -\partial G(\bar{c}; m) / \partial m \geq 0$, where $\bar{c} = t - (r - ps)\bar{q}$.

Proof. The cutoff is then type-invariant, $c^*(\theta) \equiv \bar{c}$, so $h(\theta, m) = 1 - G(\bar{c}; m)$ does not vary with θ and comes outside both integrals. The limits 0 and $\bar{\theta}$ do not depend on m , so by Leibniz's rule

$$\text{B} = [1 - G(\bar{c}; m)] \frac{d}{dm} \left[\int_0^{\bar{\theta}} f(\theta; m) d\theta \right] = [1 - G(\bar{c}; m)] \frac{d}{dm} [1] = 0,$$

since $f(\cdot; m)$ is a density for every m . Channel A collapses to $-\partial G(\bar{c}; m) / \partial m \geq 0$, because contamination shifts reporting costs up. $\square$

Why it matters. Channel B is the concealment margin: it operates only if contamination reallocates mass toward types whose exceedance risk, and hence whose cutoff, differs. Shut it down and any remaining rise in under-reporting is avoidance of the reporting technology itself. The 50%-of-MCL nitrate step is the empirical counterpart — monitoring frequency jumps, the sanction schedule does not, and readings are nowhere near the limit.

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